

Appendix to "The Method of Moderation"

Christopher D. Carroll^{a,b}, Alan Lujan^{a,b,*}, Karsten Chipeniuk^c, Kiichi Tokuoka^d, Weifeng Wu^e

^a*Johns Hopkins University, Krieger School of Arts and Sciences Baltimore, MD,*

^b*Econ-ARK,*

^c*Reserve Bank of New Zealand,*

^d*Japanese Ministry of Finance,*

^e*Fannie Mae, Washington, DC,*

Abstract

This appendix provides detailed mathematical derivations and technical results supporting the Method of Moderation. Topics include: value function transformations and their relationship to the inverse value function; explicit formulas for minimal and maximal marginal propensities to consume; cusp point calculations for tighter upper bounds; Hermite interpolation slope formulas and MPC derivations; patience conditions ensuring well-defined solutions; and extensions to stochastic returns with explicit formulas for portfolio problems.

Keywords: Dynamic Stochastic Optimization, Consumption-Saving Models, Numerical Methods

JEL classification: C63; D81; E21

1. Patience Conditions Details

Each patience condition from the main text controls a distinct way the problem could misbehave. The FVAC $0 < \beta G^{1-\rho} \mathbb{E}[\psi^{1-\rho}] < 1$ guarantees that even autarky, saving nothing and consuming income as it arrives, delivers finite expected discounted utility, so the consumer has a reason to value resources at all. The AIC $\Phi < 1$ rules out indefinite deferral of consumption: under certainty the marginal utility of consuming now exceeds the discounted marginal utility of consuming later. Two further conditions bound wealth from opposite directions. The RIC $\Phi/R < 1$ holds asset growth below the patience-adjusted discount rate, so wealth cannot explode; the GIC $\Phi/G < 1$ holds consumption growth below permanent-income growth, which is what pins down a finite target wealth ratio. Finally, the FHCW $G/R < 1$ keeps the present value of future labor income finite. Where these conditions fail, behavior changes qualitatively: when all

*Corresponding author

Email addresses: ccarroll@jhu.edu (Christopher D. Carroll), alujan@jhu.edu (Alan Lujan)

hold, the consumer runs a buffer stock around a target wealth ratio; when the AIC fails, she borrows without limit; when the GIC fails but the RIC still holds, wealth grows without bound (Carroll, 1997, 2020; Carroll and Shanker, 2024).

2. Human Wealth Formulas

The optimist's human wealth (assuming $\xi_{t+n} = 1 \forall n > 0$) can be computed three ways: backward recursion $\bar{h}_T = 0$, $\bar{h}_t = (G/R)(1 + \bar{h}_{t+1})$; forward sum $\bar{h}_t = \sum_{n=1}^{T-t} (G/R)^n$; or infinite-horizon $\bar{h} = G/(R - G)$ when $R > G$. With $G = 1$, $\bar{h} = 1/(R - 1)$.

The pessimist's human wealth (assuming $\xi_{t+n} = \underline{\xi} \forall n > 0$) follows similarly: backward recursion $\underline{h}_T = 0$, $\underline{h}_t = (G/R)(\underline{\xi} + \underline{h}_{t+1})$; forward sum $\underline{h}_t = \underline{\xi} \sum_{n=1}^{T-t} (G/R)^n$; or infinite-horizon $\underline{h} = \underline{\xi}G/(R - G)$. When $\underline{\xi} = 0$ (unemployment), $\underline{h} = 0$.

3. Marginal Propensity to Consume Formulas

The minimal MPC (perfect foresight consumer with horizon $T - t$) has three forms (Carroll, 2009): backward recursion $\underline{\kappa}_t = \underline{\kappa}_{t+1}/(\underline{\kappa}_{t+1} + \Phi/R)$ with $\underline{\kappa}_T = 1$; forward sum $\underline{\kappa}_t = (\sum_{n=0}^{T-t} (\Phi/R)^n)^{-1}$; or infinite-horizon $\underline{\kappa} = 1 - \Phi/R = 1 - (R\beta)^{1/\rho}/R$.

The maximal MPC (Carroll and Toche, 2009) satisfies backward recursion $\bar{\kappa}_t = \bar{\kappa}_{t+1}/(\bar{\kappa}_{t+1} + \wp^{1/\rho}\Phi/R)$ with $\bar{\kappa}_T = 1$; forward sum $\bar{\kappa}_t = (\sum_{n=0}^{T-t} (\wp^{1/\rho}\Phi/R)^n)^{-1}$; or infinite-horizon $\bar{\kappa} = 1 - \wp^{1/\rho}(\Phi/R)$.

4. Cusp Point Calculation

The two upper bounds intersect at the cusp point m^* where

$$\begin{aligned} (\Delta m^* + \Delta h) \underline{\kappa} &= \bar{\kappa} \Delta m^* \\ \Delta m^* &= \frac{\underline{\kappa} \Delta h}{\bar{\kappa} - \underline{\kappa}} \\ m^* &= -\underline{h} + \frac{\underline{\kappa} (\bar{h} - \underline{h})}{\bar{\kappa} - \underline{\kappa}}, \end{aligned} \tag{1}$$

where $\Delta m^* \equiv m^* - \underline{m} > 0$ since $\bar{\kappa} > \underline{\kappa}$. For $m \in (\underline{m}, m^*]$, the tighter upper bound yields

$$\begin{aligned} \Delta m \underline{\kappa} &< \hat{c}(\underline{m} + \Delta m) &< \bar{\kappa} \Delta m \\ 0 &< \hat{c}(\underline{m} + \Delta m) - \Delta m \underline{\kappa} &< \Delta m (\bar{\kappa} - \underline{\kappa}) \\ 0 &< \left(\frac{\hat{c}(\underline{m} + \Delta m) - \Delta m \underline{\kappa}}{\Delta m (\bar{\kappa} - \underline{\kappa})} \right) &< 1. \end{aligned} \tag{2}$$

This motivates the low-resource moderation ratio, defined for $m \in (\underline{m}, m^*]$ as

$$\dot{\omega}(\mu) = \frac{\hat{c}(\underline{m} + e^\mu)e^{-\mu} - \underline{\kappa}}{\bar{\kappa} - \underline{\kappa}}. \quad (3)$$

Since $e^{-\mu} = 1/\Delta m$, the right-hand side equals $(\hat{c}/\Delta m - \underline{\kappa})/(\bar{\kappa} - \underline{\kappa})$, which lies in $(0, 1)$ for $m \in (\underline{m}, m^*]$: the lower bound is the optimist's MPC and the upper bound is the maximal MPC, with strict inequality at the upper end following from $m < m^*$. Applying the logit transformation and interpolating as before yields consumption satisfying both upper bounds throughout.

5. Value Function Derivation

Under perfect foresight, consumption grows at constant rate equal to the absolute patience factor Φ : $\mathbf{c}_{t+n} = \mathbf{c}_t \Phi^n$. The present discounted value of consumption satisfies $\text{PDV}_t^T(\mathbf{c}) = \sum_{n=0}^{T-t} \beta^n \mathbf{c}_t \Phi^n = \mathbf{c}_t \sum_{n=0}^{T-t} (\Phi/R)^n$, where we use $\beta \Phi^{1-\rho} = \Phi/R$. Dividing by consumption yields the PDV-to-consumption ratio $\mathcal{C}_t^T = \text{PDV}_t^T(\mathbf{c})/\mathbf{c}_t = \sum_{n=0}^{T-t} (\Phi/R)^n = \underline{\kappa}_t^{-1}$, which is unchanged for normalized variables. Defining $\mathcal{C} \equiv \lim_{T \rightarrow \infty} \mathcal{C}_t^T$, this yields the key identity $\mathcal{C} = \underline{\kappa}^{-1}$, connecting the infinite-horizon PDV-to-consumption ratio to the minimal MPC.

The optimist's value function satisfies

$$\begin{aligned} \bar{\mathbf{v}}_{T-1}(m_{T-1}) &\equiv \mathbf{u}(c_{T-1}) + \beta \mathbf{u}(c_T) \\ &= \mathbf{u}(c_{T-1}) (1 + \beta \Phi^{1-\rho}) \\ &= \mathbf{u}(c_{T-1}) (1 + \Phi/R) \\ &= \mathbf{u}(c_{T-1}) \mathcal{C}_{T-1}^T \end{aligned} \quad (4)$$

The infinite horizon expression becomes

$$\begin{aligned} \bar{\mathbf{v}}(m) &= \mathbf{u}(\bar{\mathbf{c}}(m)) \mathcal{C} \\ &= \mathbf{u}(\bar{\mathbf{c}}(m)) \underline{\kappa}^{-1} \\ &= \mathbf{u}((\Delta m + \Delta h) \underline{\kappa}) \underline{\kappa}^{-1} \\ &= \mathbf{u}(\Delta m + \Delta h) \underline{\kappa}^{-\rho}. \end{aligned} \quad (5)$$

This can be transformed as

$$\begin{aligned} \bar{\lambda} &\equiv ((1 - \rho) \bar{\mathbf{v}})^{1/(1-\rho)} \\ &= c \mathcal{C}^{1/(1-\rho)} \\ &= (\Delta m + \Delta h) \underline{\kappa}^{-\rho/(1-\rho)}. \end{aligned} \quad (6)$$

For the realist's problem, we define $\hat{\lambda} = ((1 - \rho) \hat{\mathbf{v}}(m))^{1/(1-\rho)}$. Using the bounds $\underline{\lambda} < \hat{\lambda} < \bar{\lambda}$, we define

$$\hat{\Omega}(\mu) = \left(\frac{\hat{\lambda}(\underline{m} + e^\mu) - \underline{\lambda}(\underline{m} + e^\mu)}{\Delta h \underline{\kappa} \mathcal{C}^{1/(1-\rho)}} \right) \quad (7)$$

and the logit-transformed counterpart:

$$\begin{aligned}\hat{X}(\boldsymbol{\mu}) &= \log\left(\frac{\hat{\Omega}(\boldsymbol{\mu})}{1 - \hat{\Omega}(\boldsymbol{\mu})}\right) \\ &= \log(\hat{\Omega}(\boldsymbol{\mu})) - \log(1 - \hat{\Omega}(\boldsymbol{\mu}))\end{aligned}\tag{8}$$

Inverting these approximations yields

$$\hat{\Lambda} = \underline{\Lambda} + \overbrace{\left(\frac{1}{1 + \exp(-\hat{X})}\right)}^{=\hat{\Omega}} \Delta h \underline{\kappa} C^{1/(1-\rho)}\tag{9}$$

from which the value function approximation is $\hat{v} = u(\hat{\Lambda})$.

6. Hermite Interpolation

The numerical accuracy of the method of moderation depends critically on the quality of function approximation between gridpoints (Santos, 2000). Our bracketing approach complements work that bounds numerical errors in dynamic economic models (Judd et al., 2017). Although linear interpolation that matches the level of $\hat{c}$ at the gridpoints is simple, Hermite interpolation (Fritsch and Carlson, 1980) offers a considerable advantage.

The moderation ratio derivative measures how quickly the realist approaches the optimist as resources increase. Differentiating (??) with respect to $\boldsymbol{\mu}$ we obtain

$$\frac{\partial \omega}{\partial \boldsymbol{\mu}} = \frac{\Delta m (\partial \hat{c} / \partial m - \underline{\kappa})}{\underline{\kappa} \Delta h}.\tag{10}$$

Rearranging this yields a moderation form for the marginal propensity to consume:

$$\frac{\partial \hat{c}}{\partial m} = (1 - \boldsymbol{\eta}) \underline{\kappa} + \boldsymbol{\eta} \bar{\kappa}\tag{11}$$

where

$$\boldsymbol{\eta} = \frac{\underline{\kappa}}{\bar{\kappa} - \underline{\kappa}} \cdot \frac{\Delta h}{\Delta m} \cdot \partial \omega / \partial \boldsymbol{\mu}.\tag{12}$$

Carroll and Shanker (2024) guarantees $\underline{\kappa} \leq \partial \hat{c} / \partial m \leq \bar{\kappa}$ at gridpoints where the Euler equation holds, so $\boldsymbol{\eta} \in [0, 1]$ and the expression above is indeed a convex combination of $\underline{\kappa}$ and $\bar{\kappa}$. At very high wealth, $\boldsymbol{\eta} \rightarrow 0$ and the MPC approaches $\underline{\kappa}$; near the borrowing constraint, $\boldsymbol{\eta} \rightarrow 1$ and the MPC approaches $\bar{\kappa}$.

For Hermite interpolation, compute $\partial \omega / \partial \boldsymbol{\mu}$ at gridpoints, then derive $\partial \chi / \partial \boldsymbol{\mu} = \partial \omega / \partial \boldsymbol{\mu} / [\omega(1 - \omega)]$ for slope data. By matching both the level and the derivative of $\hat{c}$ at the gridpoints, where the derivative is obtained from the envelope condition (Benveniste and Scheinkman, 1979; Milgrom and Segal, 2002) together with

the EGM Euler equation, the interpolated consumption rule satisfies the Euler equation exactly at each solved gridpoint. These techniques extend naturally to the value function approximation.

For monotone cubic Hermite schemes (Fritsch and Carlson, 1980; Fritsch and Butland, 1984; de Boor, 2001), theoretical slopes may be adjusted to enforce monotonicity (Hyman, 1983). The Fritsch-Carlson algorithm modifies slopes at local extrema, while Fritsch-Butland uses harmonic mean weighting. Both preserve the shape-preserving property essential for consumption functions that must be strictly increasing.

7. I.I.D. Stochastic Returns: MPC derivation

The fact that a linear consumption function with an $\text{MPC} = 1 - (\beta \mathbb{E}[\mathbf{R}^{1-\rho}])^{1/\rho}$ satisfies the Euler equation with i.i.d. returns and no labor income can be derived by the method of undetermined coefficients. In particular, assume that $\bar{c}(m) = m\kappa$, with a time-independent MPC κ to be determined. Substituting this into the Euler equation, we have

$$\begin{aligned} 1 &= \beta \mathbb{E}_t \left[\mathbf{R}_{t+1} \left(\frac{c_{t+1}}{c_t} \right)^{-\rho} \right] \\ &= \beta \mathbb{E}_t \left[\mathbf{R}_{t+1} \left(\frac{m_{t+1}}{m_t} \right)^{-\rho} \right] \end{aligned} \quad (13)$$

where the second equality uses the assumed form of the consumption function. Since there is no labor income, $m_{t+1} = \mathbf{R}_{t+1}(m_t - c_t)$. Substituting this into the above we obtain

$$1 = \beta \mathbb{E}_t \left[\mathbf{R}_{t+1} (\mathbf{R}_{t+1}(1 - \kappa))^{-\rho} \right] \quad (14)$$

Solving for κ and recalling that returns are i.i.d. gives $\kappa = 1 - (\beta \mathbb{E}[\mathbf{R}^{1-\rho}])^{1/\rho}$.

In the particular case of lognormal returns, the MPC can be written in closed form. The moment generating function (MGF) for lognormal returns provides the key formula. For $\log \mathbf{R} \sim \mathcal{N}(\mu, \sigma^2)$, the MGF is $\mathbb{E}[e^{sX}] = \exp(\mu s + \sigma^2 s^2/2)$ where $X = \log \mathbf{R}$. Setting $s = 1 - \rho$ and $\mu = r + \pi - \sigma_{\mathbf{r}}^2/2$ yields¹

$$\mathbb{E}[\mathbf{R}^{1-\rho}] = \exp \left((1 - \rho) \left(r + \pi - \frac{\sigma_{\mathbf{r}}^2}{2} \right) + \frac{(1 - \rho)^2 \sigma_{\mathbf{r}}^2}{2} \right). \quad (15)$$

Simplifying the variance terms: $(1 - \rho)^2 \sigma_{\mathbf{r}}^2/2 - (1 - \rho) \sigma_{\mathbf{r}}^2/2 = (1 - \rho)[(1 - \rho) - 1] \sigma_{\mathbf{r}}^2/2 = -\rho(1 - \rho) \sigma_{\mathbf{r}}^2/2$, giving the final form

$$\mathbb{E}[\mathbf{R}^{1-\rho}] = \exp \left((1 - \rho) (r + \pi - \rho \sigma_{\mathbf{r}}^2/2) \right). \quad (16)$$

¹Here we can interpret π as the risk premium, that is, the additional average return from holding a risky asset compared to the risk-free rate r . Adjusting the average log return by the asset volatility ensures that increasing $\sigma_{\mathbf{r}}^2$ constitutes a mean-preserving spread of the level of return.

References

- Benveniste, L.M., Scheinkman, J.A., 1979. On the Differentiability of the Value Function in Dynamic Models of Economics. *Econometrica* 47, 727–732. doi:[10.2307/1910417](https://doi.org/10.2307/1910417).
- de Boor, C., 2001. A Practical Guide to Splines. volume 27 of *Applied Mathematical Sciences*. Revised ed., Springer. doi:[10.1007/978-1-4612-6333-3](https://doi.org/10.1007/978-1-4612-6333-3).
- Carroll, C.D., 1997. Buffer-Stock Saving and the Life Cycle/Permanent Income Hypothesis. *Quarterly Journal of Economics* 112, 1–55. doi:[10.1162/003355397555109](https://doi.org/10.1162/003355397555109).
- Carroll, C.D., 2009. Precautionary Saving and the Marginal Propensity to Consume Out of Permanent Income. *Journal of Monetary Economics* 56, 780–790. doi:[10.1016/j.jmoneco.2009.06.016](https://doi.org/10.1016/j.jmoneco.2009.06.016).
- Carroll, C.D., 2020. Solving microeconomic dynamic stochastic optimization problems. techreport. Johns Hopkins University. URL: <https://www.econ2.jhu.edu/people/ccarroll/SolvingMicroDSOPs.pdf>.
- Carroll, C.D., Shanker, A., 2024. Theoretical Foundations of Buffer Stock Saving (Revise and Resubmit, Quantitative Economics). URL: <https://llorracc.github.io/BufferStockTheory/BufferStockTheory.pdf>. revise and Resubmit, Quantitative Economics.
- Carroll, C.D., Toche, P., 2009. A Tractable Model of Buffer Stock Saving. Working Paper 15265. National Bureau of Economic Research. doi:[10.3386/w15265](https://doi.org/10.3386/w15265).
- Fritsch, F.N., Butland, J., 1984. A Method for Constructing Local Monotone Piecewise Cubic Interpolants. *SIAM Journal on Scientific and Statistical Computing* 5, 300–304. doi:[10.1137/0905021](https://doi.org/10.1137/0905021).
- Fritsch, F.N., Carlson, R.E., 1980. Monotone Piecewise Cubic Interpolation. *SIAM Journal on Numerical Analysis* 17, 238–246. doi:[10.1137/0717021](https://doi.org/10.1137/0717021).
- Hyman, J.M., 1983. Accurate Monotonicity-Preserving Cubic Interpolation. *SIAM Journal on Scientific and Statistical Computing* 4, 645–654. doi:[10.1137/0904045](https://doi.org/10.1137/0904045).
- Judd, K.L., Maliar, L., Maliar, S., 2017. Lower Bounds on Approximation Errors to Numerical Solutions of Dynamic Economic Models. *Econometrica* 85, 991–1020. doi:[10.3982/ecta12791](https://doi.org/10.3982/ecta12791).
- Milgrom, P., Segal, I., 2002. Envelope Theorems for Arbitrary Choice Sets. *Econometrica* 70, 583–601. doi:[10.1111/1468-0262.00296](https://doi.org/10.1111/1468-0262.00296).
- Santos, M.S., 2000. Accuracy of Numerical Solutions Using the Euler Equation Residuals. *Econometrica* 68, 1377–1402. doi:[10.1111/1468-0262.00165](https://doi.org/10.1111/1468-0262.00165).